# Quantitative Analysis of Indian IT Sector Equities

**Submitted By:** Vijay Katiyar

**Sector Chosen:** Technology Sector

**Companies Analyzed:** Infosys, TCS, Wipro

**Tools Used:** Python, Pandas, Matplotlib, Seaborn, Jupyter Notebook

## Objective

The objective of this analysis was to study the behavior of selected Indian IT sector equities using historical stock market data and identify meaningful quantitative patterns that could be useful from a trading and risk management perspective.

**For this analysis, I selected three major Indian IT companies:**

- Infosys
- Tata Consultancy Services (TCS)
- Wipro

The analysis uses statistical and quantitative techniques to study correlation, volatility, trend behavior, and the effectiveness of a simple trend-following trading strategy.

## Data Collection

Historical daily stock price data was collected using the Yahoo Finance API through Python.

The analysis used stock data from 2021 to 2024 and included:

- Closing prices

- Daily returns
- Rolling volatility
- Moving averages

Python libraries such as Pandas, Matplotlib, and Seaborn were used for data cleaning, analysis, and visualization.

## Research Methodology

The following methodology was used during the analysis:

1. Historical daily stock data was collected for Infosys, TCS, and Wipro.
2. Daily percentage returns were calculated using closing prices.
3. Correlation and volatility metrics were analyzed.
4. Rolling moving averages were generated to identify trend behavior.
5. A moving average crossover strategy was created.
6. The strategy was backtested against a buy-and-hold benchmark.
7. Performance and risk-adjusted return metrics were evaluated.

## Correlation Analysis

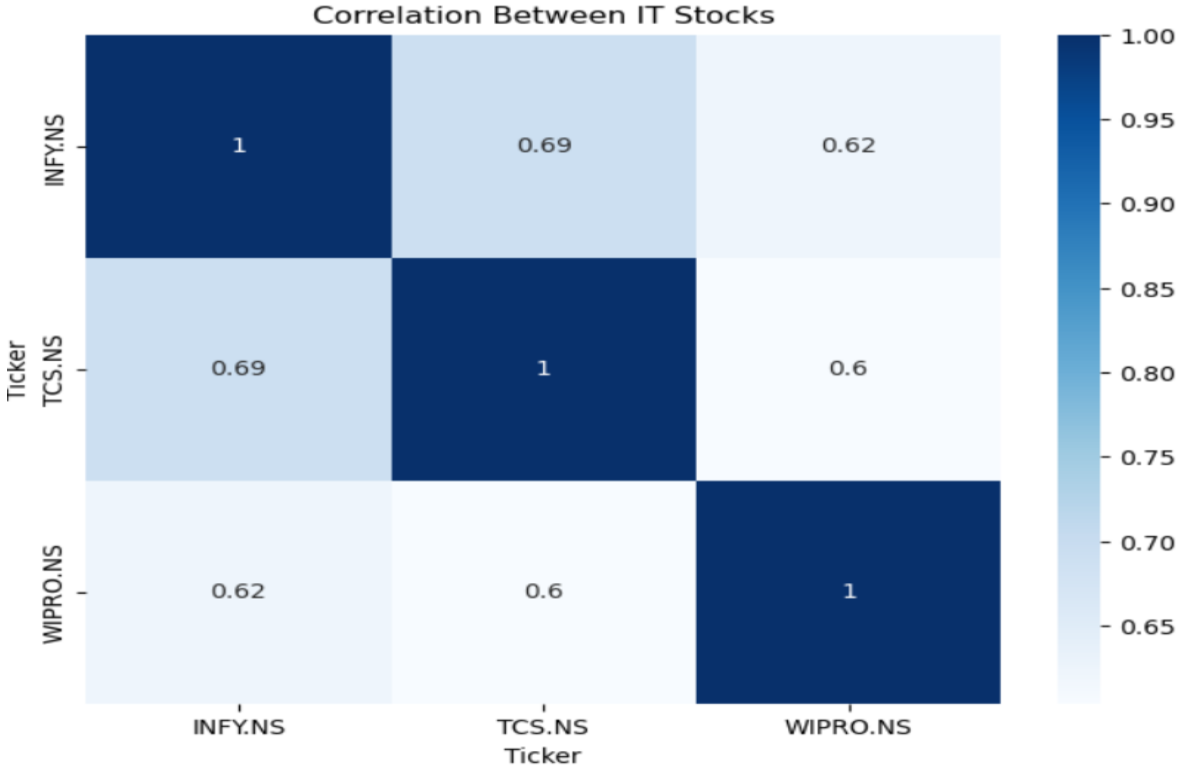


Figure 1: Correlation Matrix of Daily Returns

The first major observation was the strong positive correlation between Infosys, TCS, and Wipro.

The correlation matrix showed that the daily returns of these companies moved similarly during most market conditions. Infosys and TCS showed particularly high correlation, while Wipro also followed similar market trends.

This indicates that large IT companies in India are influenced by similar macroeconomic and sector-specific factors such as:

- global technology demand,
- US market sentiment,
- currency fluctuations,
- and investor expectations.

One important implication is that diversification within the same sector may provide limited risk reduction during market-wide movements.

## Volatility Analysis

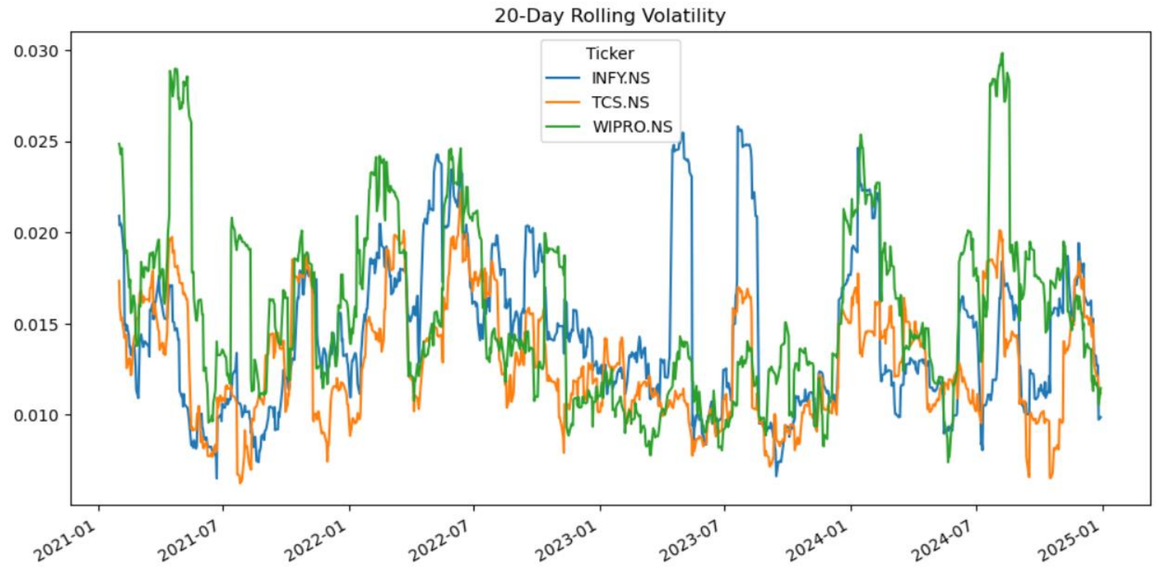


Figure 2: Rolling Volatility of IT Stocks

The rolling volatility analysis showed that volatility increased significantly during uncertain market periods.

Standard deviation of returns was used as a measure of market volatility and risk. During periods of global economic uncertainty, all three stocks experienced higher price fluctuations and larger daily return swings.

This suggests that Indian IT equities are highly sensitive to changes in global technology sentiment and macroeconomic conditions.

Higher volatility may increase trading opportunities, but it also increases overall market risk exposure.

## Trend Analysis Using Moving Averages

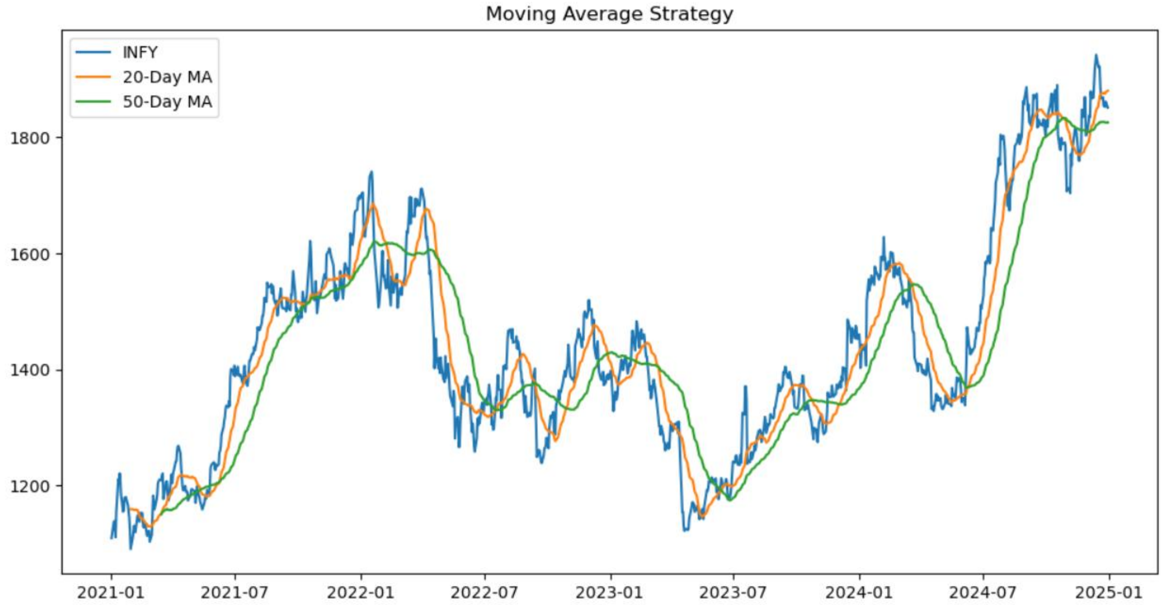


Figure 3: Moving Average Trend Analysis for Infosys

To study market trends, 20-day and 50-day moving averages were applied to Infosys stock prices.

The analysis showed that during strong upward market trends, the 20-day moving average remained above the 50-day moving average for extended periods. Similarly, bearish market phases were visible when the shorter moving average moved below the longer moving average.

This indicated that moving averages can help identify medium-term market momentum and trend direction.

## Trading Idea

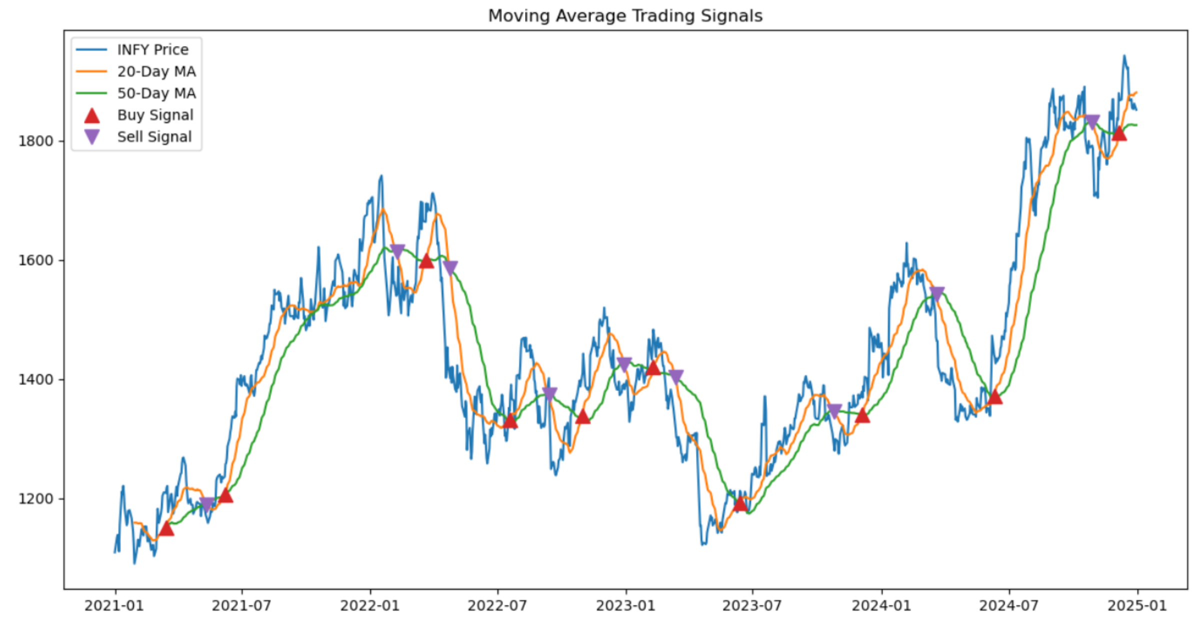


Figure 4: Moving Average Crossover Trading Signals

Based on the observed trend-following behavior, a simple moving average crossover strategy was tested.

Trading Rules:

- Buy when the 20-day moving average crosses above the 50-day moving average.

- Sell when the 20-day moving average crosses below the 50-day moving average.

Hypothesis:

If medium-term trends exist in IT equities, then moving average crossover signals may help identify directional momentum more effectively than random market entry points.

The strategy was designed to provide a rule-based approach for identifying market trends while reducing emotional trading decisions.

## Backtesting the Strategy

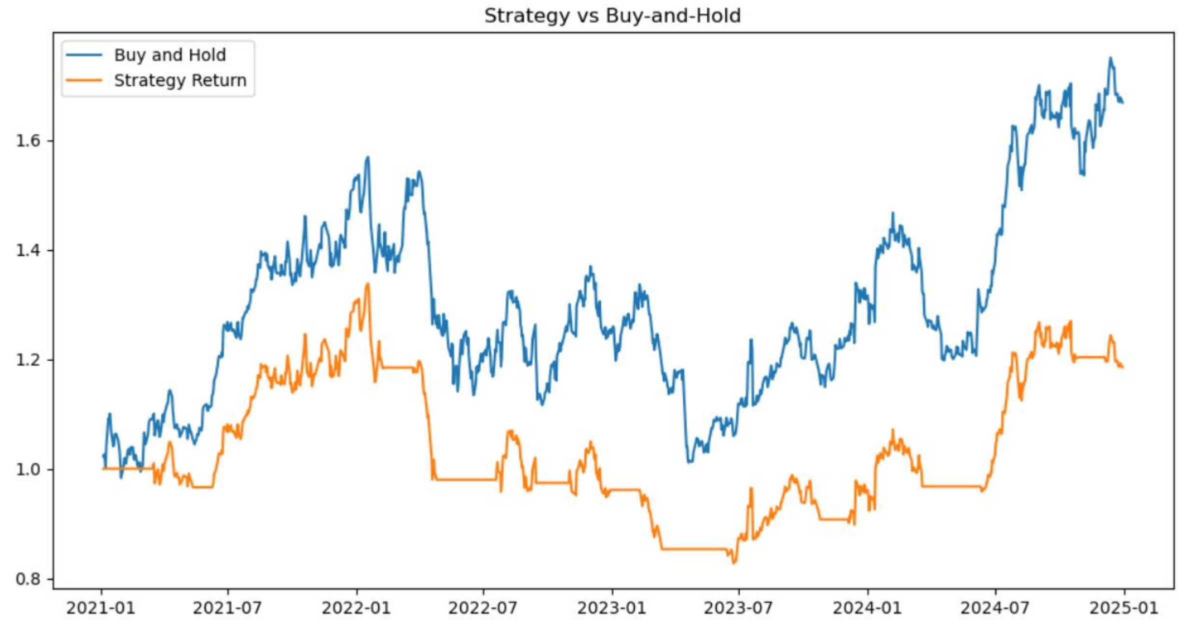


Figure 5: Strategy Performance vs Buy-and-Hold

The strategy was backtested using historical price data and compared against a simple buy-and-hold benchmark.

The purpose of the backtest was to evaluate whether the trading signals provided useful trend identification beyond random market movement.

The strategy was able to capture several major upward market trends while avoiding some weaker market phases. Although the strategy did not outperform the benchmark during all periods, it demonstrated the usefulness of systematic and rule-based trading approaches.

## Performance Metrics

The following performance metrics were calculated:

- Annual Return

- Annual Volatility

- Sharpe Ratio

These metrics were used to evaluate both profitability and risk-adjusted performance.

The Sharpe Ratio provided insight into how efficiently the strategy generated returns relative to the level of risk taken.

**Annual Return: 0.05995961038853734**

**Annual Volatility: 0.1812110518107012**

**Sharpe Ratio: 0.3308827457785138**

The strategy generated moderate positive returns with controlled volatility, suggesting that trend-following techniques may be useful in strongly directional markets.

## Risks and Limitations

There are several limitations associated with this approach:

- Moving averages are lagging indicators and may react slowly during sudden market reversals.

- False buy and sell signals may occur during sideways or highly volatile market conditions.

- The strategy relies only on historical price data and does not consider company fundamentals or news events.

- Transaction costs and slippage may reduce actual profitability in real trading conditions.

- The model does not estimate the probability distribution of extreme market events, which may lead to unexpected drawdowns during highly volatile periods.

Therefore, the strategy should be considered as a basic quantitative model rather than a complete trading system.

## Use of AI

AI tools were used during the initial brainstorming and structuring phase of this report.

However, the data collection, coding, chart generation, analysis, interpretation of results, and validation of observations were performed manually by me.

## Conclusion

This analysis helped in understanding how major Indian IT sector equities behave under different market conditions using historical market data.

The study showed that:

- IT stocks are highly correlated,
- volatility increases during uncertain periods,
- and moving average indicators can provide useful trend-following signals.

Although the strategy has limitations, the project demonstrated how quantitative and statistical techniques can be applied to generate practical market insights and evaluate trading ideas using historical data.